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REPUBLIC POLYTECHNIC

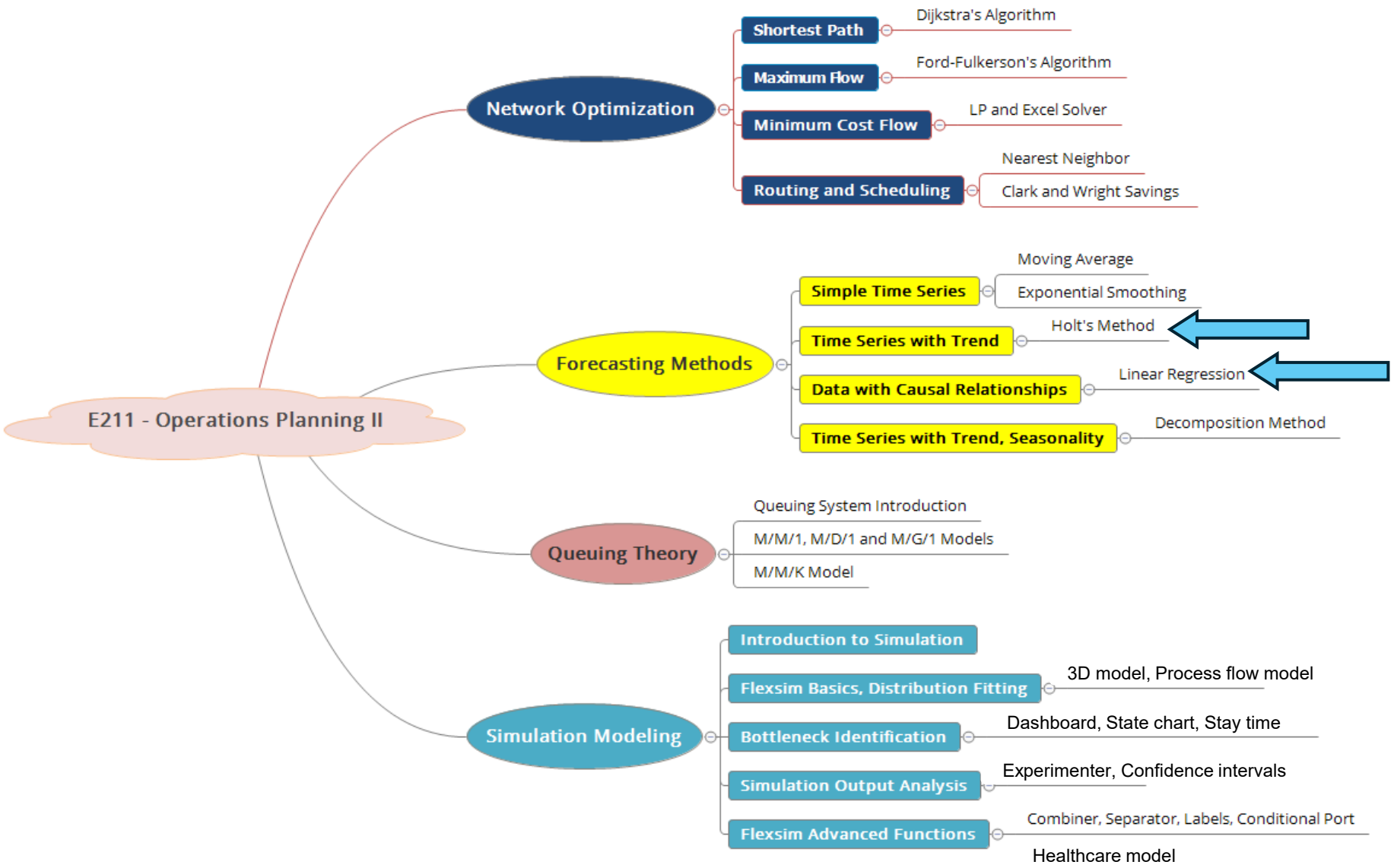
Lesson 23

Demand Forecasting: Linear Regression & Holt's Method

E211 – Operations Planning II

REPUBLIC POLYTECHNIC

E211 Topic Tree



Time Series Forecasting Methods

1. Moving Average (MA)
2. Weighted Moving Average (WMA)
3. Simple Exponential Smoothing (SES)
4. Trend-adjusted exponential smoothing (Holt's method)
5. Decomposition method
    1. Multiplicative decomposition method
    2. Additive decomposition method

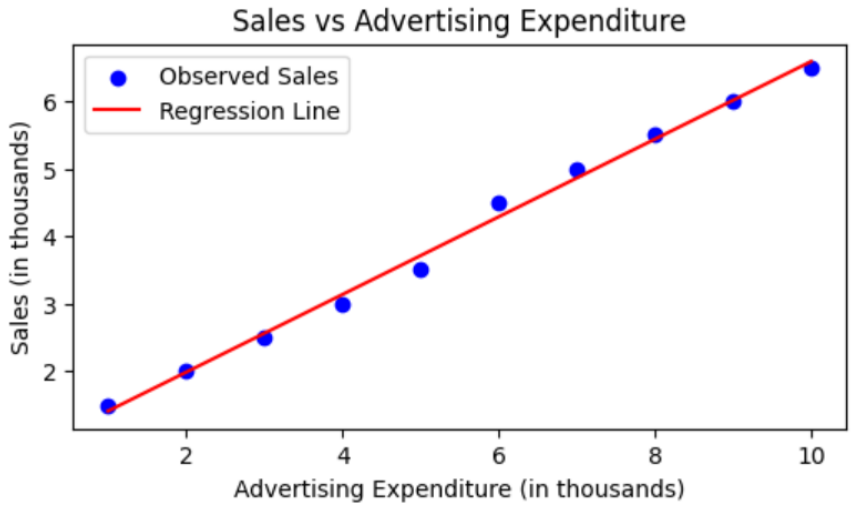
We will also cover Linear Regression, which is a causal forecasting method, not time series.

Linear Regression

Simple Linear Regression

- The simple linear regression model assumes that the relationship between the dependent (response) variable, which is denoted by y, and the independent (predictor) variable, denoted by x, can be approximated by a straight line.

- Examples of simple linear regression model:
  - Forecasting price of house (y) from size of house (x)
  - Forecasting sales of product (y) from advertising spending (x)



Simple Linear Regression

- Relates dependent variable to one independent variable in the form of a linear equation

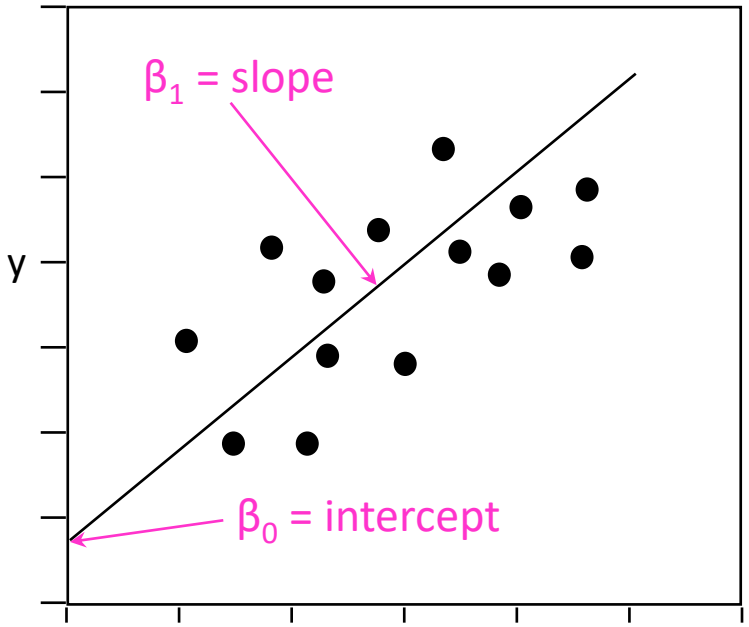
Dependent variable

Independent variable

$$y = \beta_0 + \beta_1 x$$

Unknown parameters to be estimated from data

$\beta_1 > 0 \Rightarrow$ **Positive Association**
$\beta_1 < 0 \Rightarrow$ **Negative Association**
$\beta_1 = 0 \Rightarrow$ **No Association**



The Linear Regression model **minimizes** the sum of squared differences between the points and the line (difference between actual and predicted value).

Regression Model Statistical Measure – r and R^2

Correlation coefficient, r value is between -1 and 1.

- **r measures the strength of the relationship**. If r is close to 1 or -1, it indicates that there is a strong relationship between the dependent and independent variables.
- If r is positive, β_1 (slope) is positive and vice versa. The sign of r indicates whether the variables are positively (+ sign) or negatively (- sign) correlated.

Coefficient of determination, R^2 value is between 0 and 1.

- **R^2 measures how well the regression model fits the data**.
- R^2 indicates the proportion of variation in the dependent variable that can be explained by the variation in the independent variable.
- The rest of the variations are due to other factors not accounted for in the model.

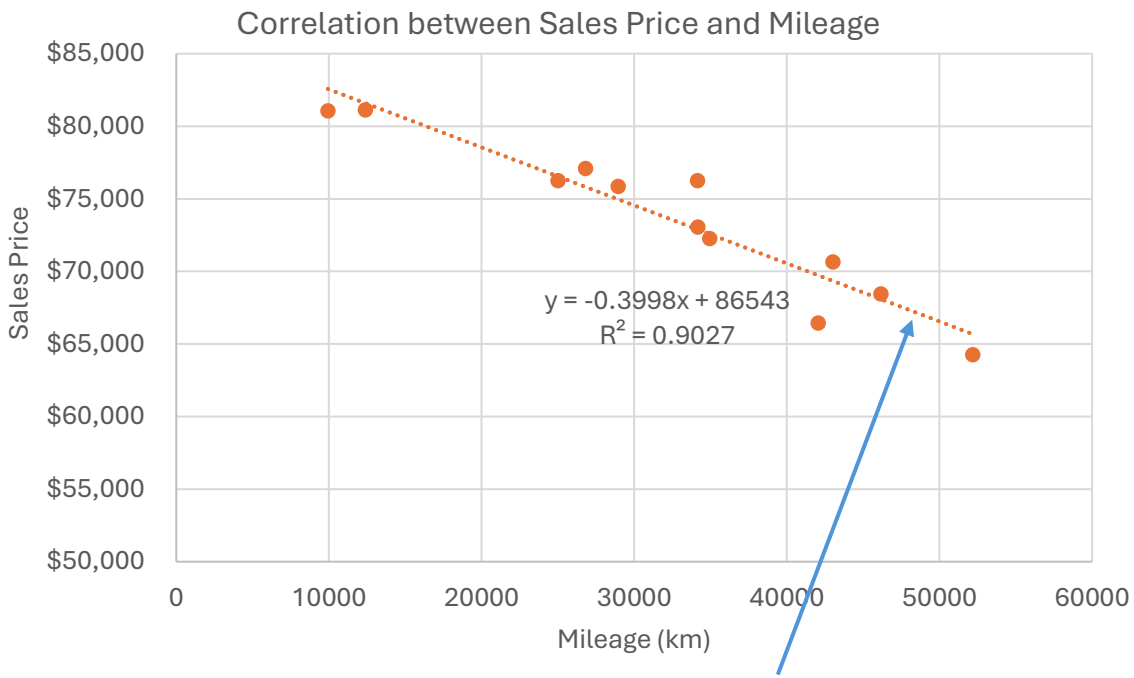
Linear Regression Exercise

Anson is the Sales Manager of Fleet Motors which sells 3-year-old previously leased cars of the same made. Anson is interested to study the relationship between the mileage (distance vehicle has travelled since manufactured) and the sales price of these vehicles. The following table shows the sales price and mileage data for 12 vehicles recently sold:

Sales Price (SGD)	Mileage (km)
$81,048	9927
$72,248	34947
$64,258	52178
$77,088	26817
$66,448	42047
$76,248	34147
$76,248	25003
$73,048	34159
$70,648	43017
$81,128	12378
$68,448	46168
$75,848	28947

1. Identify the dependent and independent variables.
2. Use a Scatter Plot to obtain a linear regression model.
3. State β_0 and β_1.
4. Comment on the linear relationship between Sales Price and Mileage and interpret the R^2 value.
5. Predict the sales price for a 3-year-old car with mileage of 20000 km.

Linear Regression Exercise



Add 'Trendline' (linear) to get the best-fit line

1. Dependent variable (Y) is sales price. Independent variable (X) is mileage.

2. Add a linear trendline to Scatter Plot to obtain the linear regression model.

3. β_0 = 86543 (intercept). β_1 = -0.3998 (slope)

4. r = square root of R^2. correlation coefficient r is -0.9501 since the slope is negative. It shows that there is a strong negative correlation between sales price and mileage. As mileage increases, sales price will decrease. Coefficient of determination R^2 = 0.9027. This means that 90.27% of the variation in sales price can be explained by the variation in mileage.

5. Linear regression model Y = -0.3998*x + 86543. Subst. x=20000, Y = $78547.

Slope/ Intercept of linear trend line, can be determined by:

- Scatter Plot ('Linear trendline')
- Excel function: Slope(known_ys, known_xs), Intercept (known_ys, known_xs)

Holt’s Method

Holt's Method

- Also known as Holt's Linear Trend Method or Double Exponential Smoothing
- Holt's method is a two-parameter exponential smoothing model α and β
- It extends simple exponential smoothing to include a linear-trend component. Contains two forecast components:
  - **Time series level L_t**
  - **Time series trend T_t**

Holt's Method

Lt = Intercept, Tt = Slope (trend) for equation (a)

- The Holt's Model comprises of 3 equations below:

$$L_{t+1} = \alpha A_{t+1} + (1-\alpha)(L_t + T_t) \quad \text{------- a}$$

$$T_{t+1} = \beta(L_{t+1} - L_t) + (1-\beta)T_t \quad \text{------- b}$$

$$H_{t+m} = L_t + mT_t \quad \text{---------- c}$$

where equation:

a is the <u>estimate of the time series' level</u>

b is the <u>estimate of the time series' trend</u>

c is the <u>overall forecast</u>

L_{t+1} : estimate of time series' level for period *t* +1

A_{t+1} : actual value of time series at period *t* +1

L_t : estimate of time series' level for period *t*

T_t : estimate of time series' trend for period *t*

$\alpha(\text{Alpha})$: smoothing constant for the level estimate; $\alpha \in [0, 1]$

$\beta(\text{Beta})$: smoothing constant for the trend estimate; $\beta \in [0, 1]$

m : number of periods ahead to be forecasted

H_{t+m} : Holt's forecast value for period *t* + *m*

Selection of Smoothing Constants

- When small values are used, less weight is given to historical data, effectively smoothing out random noise. This is appropriate under stable conditions.
- If the level and/or trend of the underlying data are changing appreciably over time, then higher values of the smoothing constants are suitable for quicker responses to level and/or trend changes.
- Because one can never really be sure if forecasting errors are a result of random noise or a real change, some guesswork (trial and error) may be required to produce a suitable model.

Exercise

Zada is a university student majoring in sociology. In her final year project, she is planning to forecast the percentage of senior citizen (65 years old and above) in her country for the next ten years.

Data (table below) were collected for the past six decades (one data sampled for every five years). Zada noticed that the numbers have been increasing over time with a visible trend. However, the rate of increase may not be the same over the years, and rate of increase in the recent years could be more indicative of future forecasts.

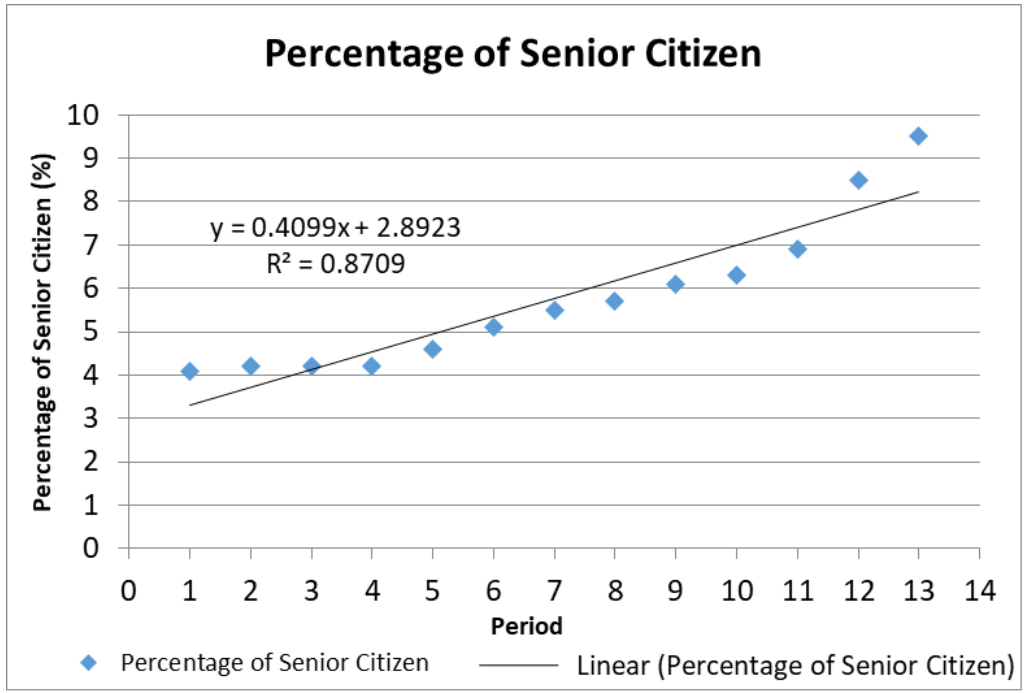
Year	Percentage of Senior Citizen
1960	4.1
1965	4.2
1970	4.2
1975	4.2
1980	4.6
1985	5.1
1990	5.5
1995	5.7
2000	6.1
2005	6.3
2010	6.9
2015	8.5
2020	9.5
2025	???
2030	???

1. Forecast the percentage of senior citizen for the next ten years (Year 2025 and 2030) using
   - Linear Regression method
   - Holt's method ($\alpha = 0.7$, $\beta = 0.2$)

2. Explain why Holt's method performs better than Linear Regression.

Exercise – Linear Regression Model

Year	Period	Percentage of Senior Citizen	Linear regression Forecast
1960	1	4.1	3.3022
1965	2	4.2	3.7121
1970	3	4.2	4.1220
1975	4	4.2	4.5319
1980	5	4.6	4.9418
1985	6	5.1	5.3516
1990	7	5.5	5.7615
1995	8	5.7	6.1714
2000	9	6.1	6.5813
2005	10	6.3	6.9912
2010	11	6.9	7.4011
2015	12	8.5	7.8110
2020	13	9.5	8.2209
2025	14	???	8.6308
2030	15	???	9.0407



Linear regression model:

$y = 0.4099x + 2.8923$

$y_{14} = 0.4099*14 + 2.8923 = 8.6308$

$y_{15} = 0.4099*15 + 2.8923 = 9.0407$

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Exercise – Holt's Method

Year	Period	Percentage of Senior Citizen	L_t	T_t	Holt's H_t
			2.8923	0.4099	
1960	1	4.1	3.8607	0.5216	3.3022
1965	2	4.2	4.2547	0.4961	4.3822
1970	3	4.2	4.3652	0.4190	4.7507
1975	4	4.2	4.3753	0.3372	4.7842
1980	5	4.6	4.6337	0.3214	4.7124
1985	6	5.1	5.0566	0.3417	4.9552
1990	7	5.5	5.4695	0.3560	5.3983
1995	8	5.7	5.7376	0.3384	5.8254
2000	9	6.1	6.0928	0.3418	6.0760
2005	10	6.3	6.3404	0.3229	6.4346
2010	11	6.9	6.8290	0.3561	6.6633
2015	12	8.5	8.1055	0.5401	7.1850
2020	13	9.5	9.2437	0.6598	8.6457
2025	14	???			9.9035
2030	15	???			10.5632

T_0: Slope of linear trend line

L_0: intercept of linear trend line

$$L_{t+1} = \alpha A_{t+1} + (1-\alpha)(L_t + T_t) \quad \text{------- a}$$

$$T_{t+1} = \beta(L_{t+1} - L_t) + (1-\beta)T_t \quad \text{------- b}$$

$$H_{t+m} = L_t + mT_t \quad \text{---------- c}$$

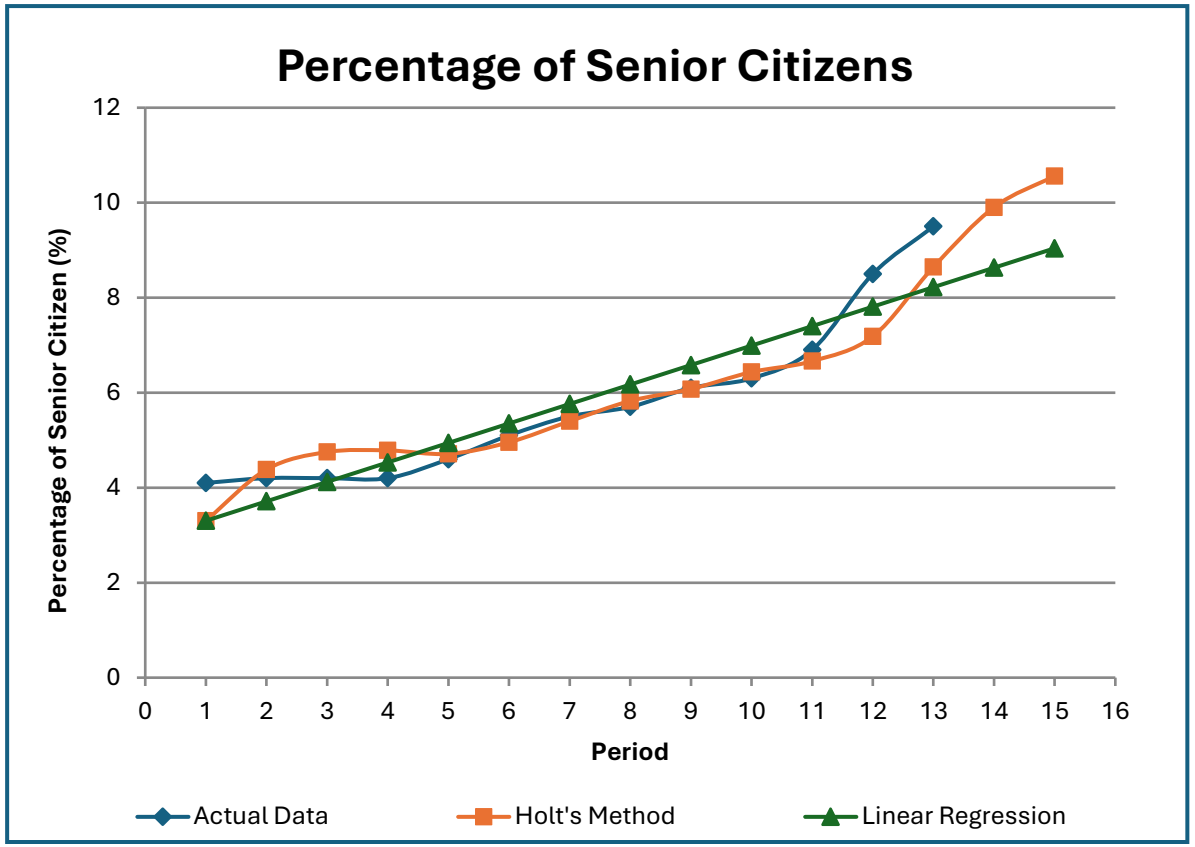
$H_{14} = L_{13} + T_{13}$

$H_{15} = L_{13} + 2*T_{13}$

$L_{12} = 0.7*A_{12} + (1-0.7)*(L_{11}+T_{11})$

$T_{12} = 0.2*(L_{12}-L_{11}) + (1-0.2)*T_{11}$

Exercise – Comparison



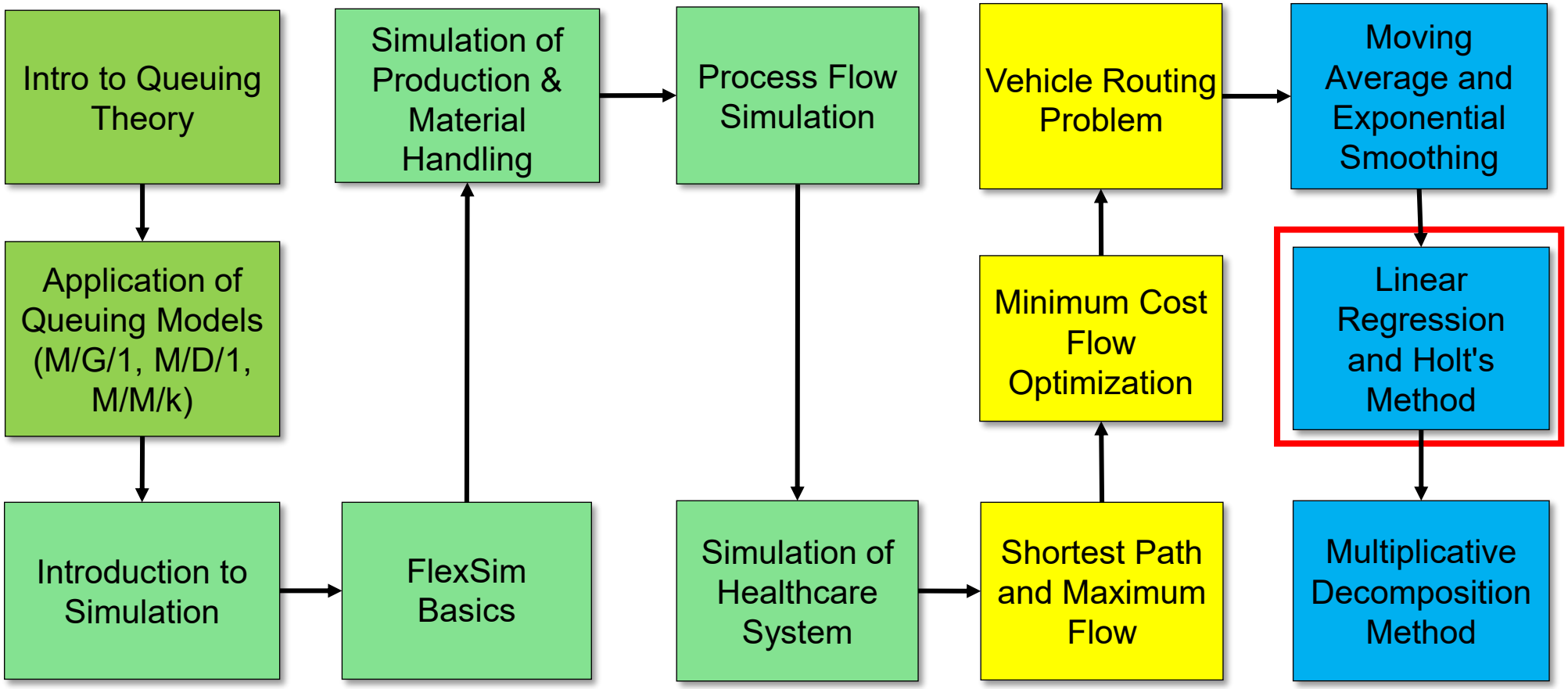
Forecasting methods	Measures of forecasting error	
	MAD	MAPE
Linear Regression	0.51	8.76%
Holt's (α = 0.7, β = 0.2)	0.40	6.97%

Holt's Method with α = 0.7 and β = 0.2 gives a more accurate forecast due to having smaller forecasting errors. The better performance of Holt's Method over Linear Regression is because:

- The rate of increase in the historical data is not constant over the time periods.
- Linear Regression treats all data in the time series equally.
- Holt's Method can put more weight on more recent data and therefore can better capture the recent change in the data.

Learning Outcomes

At the end of the lesson, students should be able to:

- Apply Linear Regression on historical data to determine whether there is any linear relationship between two factors.
- Perform a forecast from the regression line.
- Apply the Holt's method to forecast time series with trend component.

E211 Progression